

ERP Sales & Inventory Transaction Model

Phase 1 Baseline v2 — Sales Order, Inventory Allocation & Customer Aggregates

Purpose. This updated baseline incorporates the agreed policy controls and implementation decisions before Phase 1 development.

1. Transaction Lifecycle

Sales Order → Delivery Receipt → Invoice → Payment Receipt

Document	Business purpose	Inventory effect	Financial effect
Sales Order	Customer commitment	Allocate quantity	Total Orders only
Delivery Receipt	Physical fulfillment	On-Hand – delivery; Allocated – delivery; Delivered – delivery	Revenue
Invoice	Revenue recognition	No physical movement	Total Sales +; receivable increases
Payment Receipt	Settlement	No physical movement	Total Receipts +; receivable decreases

2. Customer Aggregates

Field	Storage	Definition
Total Orders	Stored	Sum of Sales Order amounts
Total Sales	Stored	Sum of Invoice amounts / recognized sales
Total Receipts	Stored	Sum of payment receipts
Receivable	Derived	Total Sales – Total Receipts; do not store

The existing Sales Order aggregate pattern is redirected from Customers.Total Sales to Customers.Total Orders. Invoice logic will later own Total Sales.

3. Approved Inventory Schema

Field	Definition
Item ID	Primary key
Item Type / Category / Subcategory	Classification
Item Name	Item description
QTY On-Hand	Physical stock
QTY Allocated	Committed to outstanding Sales Orders
QTY Delivered	Cumulative quantity physically delivered
QTY Available	Derived: QTY On-Hand – QTY Allocated
QTY On-Order	Supplier quantity ordered but not received
Reorder Level	Procurement threshold
Item Status	Active / Inactive / Discontinued

4. Configurable Over-Allocation Policy

ALLOW_OVER_ALLOCATION = TRUE/FALSE

When TRUE, an SO may allocate beyond physical On-Hand and QTY Available may become negative. This is treated as a procurement signal, not an error. When FALSE, the system prevents an allocation that would exceed On-Hand.

Regardless of the setting, the UI should warn the user when the proposed allocation would create negative availability. The warning informs the user; the configuration determines whether saving is permitted.

5. Inventory Rules

Event	Effect
Create SO	Inventory.QTY Allocated += ordered quantity
Edit SO	Apply the allocation delta between old and new business state
Delete / cancel SO	Release outstanding allocation
Delivery Receipt	On-Hand -= delivered; Allocated -= delivered; Delivered += delivered
Future Purchase Receipt	On-Order -= received; On-Hand += received

6. Sales Order Edit / Allocation Reconciliation

Allocation changes are a Sales-domain business rule. The generic Document Engine remains responsible for document/detail persistence and reconciliation; it should not own inventory policy.

The Sales Order service determines the business delta and applies inventory changes within the same transaction/lock boundary as the SO save.

Explicit cases: quantity increase, quantity decrease, detail added, detail removed, item changed on an existing detail, customer changed, and SO deletion/cancellation.

7. Phase 1 Implementation Sequence

1. Update Sales Order customer aggregate: Total Orders replaces Total Sales for SO transactions.
2. Implement Inventory allocation state and derived QTY Available.
3. Implement configurable over-allocation policy and UI shortage warning.
4. Implement SO allocation reconciliation for create, edit, customer reassignment, detail changes, and deletion.
5. Test allocation math and customer aggregates before building downstream documents.
6. Only after Phase 1 is stable, implement Delivery Receipt.
7. Then implement Invoice as the revenue-recognition event.
8. Then implement Payment Receipt as receivable settlement.

8. Architectural Principles

Keep generic persistence in the Repository / Document Engine, Sales-specific rules in the Sales domain/service, and inventory rules in the Inventory domain/service. Cross-domain updates must execute within the same transaction/lock boundary so an SO cannot be saved without its corresponding allocation update.

Derived values should be calculated rather than stored when their inputs are already persisted: Receivable = Total Sales – Total Receipts; QTY Available = QTY On-Hand – QTY Allocated.

9. Phase 1 Success Criteria

- ✓ Sales Orders are commitments, not recognized sales.
- ✓ Sales Orders update Customer.Total Orders, not Total Sales.
- ✓ SO creation allocates inventory without deducting physical stock.
- ✓ Over-allocation is controlled by a configuration switch.
- ✓ Negative availability produces a visible shortage warning.
- ✓ SO edits correctly reconcile allocation deltas, including item replacement.
- ✓ SO deletion/cancellation releases outstanding allocation.
- ✓ Receivable is derived, not stored.
- ✓ Phase 1 is validated before Delivery Receipt implementation begins.